

Phone Book Management System Specification

Version 0.1

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Bảng ghi nhận thay đổi tài liệu

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21/08/26	0.1	xây dựng tài liệu	Đặng Quốc Tấn

I. Functional Requirements The Phone Book Management System needs to be built with the following core features:

1. Contact Management:

- + Users can add a new contact by providing necessary information including: Full Name, Phone Number, Email, Address, and Group/Category.
- + Users can view the full list of contacts. The list can be sorted alphabetically by name (A-Z or Z-A).
- + Users can select a specific contact to view detailed information.
- + Users can update (edit) or delete an existing contact from the system.

2. Search and Filter:

- + Users can search for a contact by inputting a keyword (Name or Phone Number). The system will return a list of contacts matching the keyword.
- + Users can filter contacts by their assigned Group/Category (e.g., Family, Friends, Work).

3. Group/Category Management:

- + Users can create new groups to classify their contacts.
- + Users can edit the name of an existing group or delete a group. (If a group is deleted, the contacts within it will be moved to a default "Unassigned" group).

4. User Authentication (Account Management):

- + The system requires users to log in with a username/email and password to access their personal phone book.
- + New users can register for an account by providing basic personal information and a secure password.
- + Users can change their password or update their profile information.

II. Glossary

1. **Contact:** An individual entry in the phone book containing personal details such as name, phone number, and email.
2. **Group/Category:** A user-defined label used to organize and classify contacts (e.g., VIP, Family, Colleagues).
3. **Keyword:** A string of characters (letters or numbers) inputted by the user to search for a specific contact.

III. Supplementary Specification

- + **Usability:** The user interface must be intuitive and easy to navigate. If it is a console application, the menu options must be clearly presented with text instructions. Error messages must be clear and guide the user on how to fix the input.
- + **Reliability:** The system must ensure data persistence. Contact data must not be lost in the event of an unexpected application crash or sudden exit.
- + **Performance:** The system must be capable of storing at least 10,000 contacts without noticeable lag. Search and filter operations must return results in less than 1 second.
- + **Supportability:** The source code must be well-documented and highly modular to allow easy addition of new features (e.g., importing/exporting contacts) in the future.
- + **Security:** User passwords must be encrypted (hashed) before being stored in the database. The system must implement input validation to prevent SQL Injection and handle invalid data entries (e.g., entering letters in the phone number field).
- + **Design Constraints:** The system will be developed using Python. The database schema will be managed using MySQL.

